

Grad Cafe: SQL analysis

Module 3 – Wendy Eloë

Question 1: How many entries are for Fall 2026?

Result: 31,914

SQL: `SELECT COUNT(*) FROM applicants WHERE lower(trim(term)) = 'fall 2026';`

Explanation: This counts entries with the specified term, ignoring capitalization and surrounding spaces.

Question 2: Among entries with a nationality classification, what percentage are international?

Result: 45.70%

SQL: `SELECT 100.0 * COUNT(*) FILTER (WHERE lower(trim(us_or_international)) = 'international') / NULLIF(COUNT(NULLIF(trim(us_or_international), '')), 0) FROM applicants;`

Explanation: This counts entries with a nationality of international (ignoring capitalization and surrounding spaces) and divides that by the count of nonblank nationalities that aren't international.

Question 3: What are the average GPA, GRE Quantitative, GRE Verbal, and GRE Analytical Writing scores?

Result: Average GPA: 3.77; Average GRW Quantitative: 165.79; Average GRE Verbal: 160.73; Average GRE Analytical Writing: 4.35

SQL: `SELECT AVG(gpa), AVG(gre), AVG(gre_v), AVG(gre_aw) FROM applicants;`

Explanation: AVG ignores NULL separately for each metric, so an entry can contribute to one average without providing the others.

Question 4: What is the average GPA of American applicants for Fall 2026?

Result: 3.79

SQL: `SELECT AVG(gpa) FROM applicants WHERE lower(trim(term)) = 'fall 2026' AND lower(trim(us_or_international)) = 'american';`

Explanation: Restrict the term and nationality, then average the available GPAs.

Question 5: What percentage of Fall 2025 entries are acceptances?

Result: 47.23%

SQL: `SELECT 100.0 * COUNT(*) FILTER (WHERE lower(trim(status)) IN ('accepted', 'acceptance', 'admitted')) / NULLIF(COUNT(*), 0) FROM applicants WHERE lower(trim(term)) = 'fall 2025';`

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Explanation: Divide accepted Fall 2025 entries by all Fall 2025 entries. A zero denominator produces N/A.

Question 6: What is the average GPA of accepted applicants for Fall 2026?

Result: 3.76

SQL: `SELECT AVG(gpa) FROM applicants WHERE lower(trim(term)) = 'fall 2026' AND lower(trim(status)) IN ('accepted', 'acceptance', 'admitted');`

Explanation: Restrict the term and admission status, then average available GPAs.

Question 7: How many entries are for a Computer Science master's degree at Johns Hopkins University (original fields)?

Result: 8

SQL: `SELECT COUNT(*) FROM applicants WHERE program ~* '(johns hopkins|mjh|M)' AND program ~* 'computer\s+science' AND lower(trim(degree)) IN ('masters', 'master', 'master's', 'ms', 'm.s.', 'msc', 'm.sc.', 'meng', 'm.eng.');`

Explanation: Match Computer Science and Johns Hopkins or the JHU acronym in the original combined program field, and require a master degree variant.

Question 8: How many Fall 2026 acceptances are for a Computer Science PhD at Georgetown, MIT, Stanford, or Carnegie Mellon (original fields)?

Result: 30

SQL: `SELECT COUNT(*) FROM applicants WHERE lower(trim(term)) = 'fall 2026' AND lower(trim(status)) IN ('accepted', 'acceptance', 'admitted') AND lower(trim(degree)) IN ('phd', 'ph.d.', 'ph.d', 'doctor of philosophy') AND program ~* 'computer\s+science' AND program ~* '(georgetown|massachusetts institute of technology|mmit|M|stanford|carnegie mellon|mcmu\M)';`

Explanation: Require all five conditions: Fall 2026, acceptance, PhD, Computer Science, and one of the four named universities.

Question 9: How does the Question 8 count change when using LLM-generated university and program fields?

Result: Original-field count: 30; LLM-field count: 30; Difference: +0

SQL: `SELECT COUNT(*) FROM applicants WHERE lower(trim(term)) = 'fall 2026' AND lower(trim(status)) IN ('accepted', 'acceptance', 'admitted') AND lower(trim(degree)) IN ('phd', 'ph.d.', 'ph.d', 'doctor of philosophy') AND llm_generated_program ~* 'computer\s+science' AND llm_generated_university ~* '(georgetown|massachusetts institute of technology|mmit|M|stanford|carnegie mellon|mcmu\M)';`

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Explanation: Repeat Q8 with the standardized university and program fields; subtract the original count from the LLM count. Original term, status, and degree still control eligibility. Both counts are 30 in this snapshot, giving a difference of +0. The normalization did not change the total number of matching entries. Equal counts alone do not prove identical membership or accurate names. In other snapshots, expanded acronyms, corrected spellings, or mistaken LLM standardizations could add or remove matches.

Question 10: Which five programs have the highest average GRE Quantitative scores (at least 10 valid scores)?

Result: finance: 168.39 (23 scores); economics: 167.91 (400 scores); mechanical engineering: 167.39 (23 scores); chemistry: 167.39 (18 scores); statistics: 165.65 (17 scores)

SQL:

```
SELECT lower(trim(llm_generated_program)) AS program, AVG(gre) AS average_gre, COUNT(gre) AS score_count FROM applicants WHERE gre BETWEEN 130 AND 170 AND lower(trim(llm_generated_program)) NOT IN ('', 'unknown', 'n/a') GROUP BY lower(trim(llm_generated_program)) HAVING COUNT(gre) >= 10 ORDER BY average_gre DESC, program ASC LIMIT 5;
```

Explanation: Group standardized program names across all terms and universities, ignoring case and surrounding spaces. Exclude missing names and invalid GRE Quantitative scores, require at least 10 scores, and rank by average score. Alphabetical order breaks ties.

Question 11: Which five universities have the most reported rejections across all terms?

Result: stanford university: 528 rejections; yale university: 451 rejections; princeton university: 413 rejections; university of california, berkeley: 395 rejections; massachusetts institute of technology: 373 rejections

SQL:

```
SELECT lower(trim(llm_generated_university)) AS university, COUNT(*) AS rejection_count FROM applicants WHERE lower(trim(status)) = 'rejected' AND lower(trim(llm_generated_university)) NOT IN ('', 'unknown', 'n/a') GROUP BY lower(trim(llm_generated_university)) ORDER BY rejection_count DESC, university ASC LIMIT 5;
```

Explanation: Count rejected entries across all terms by standardized university name, ignoring case and surrounding spaces and excluding missing names. Return the five largest counts, breaking ties alphabetically. These are reported counts, not rejection rates.